

FRONT END DEVELOPMENT FRAMEWORKS AND UI ENGINEERING (25CS1201E)
2025-26, Term-III
Project Problem Statement Submission Form

Sec No-	Team No-	Project Title-
Team Lead Roll Number		2520090139
Team Lead Name		A. Shashank Reddy
Team Member Roll Number		2520030087
Team Member Name		P. Anudeep

1. Abstract

The Online Examination Platform is a web-based system designed to make conducting exams easier, faster, and more efficient. Traditional exam methods are time-consuming, require manual grading, and can be difficult to manage, especially in online or large-scale learning environments. There is also a growing need for systems that can provide quick results while maintaining fairness and reliability.

This project offers a simple solution by allowing teachers to create exams with multiple-choice and essay questions through an easy-to-use exam builder. The platform automatically grades objective questions, saving time and reducing human error, while subjective answers can be reviewed manually. Features like a timer and basic anti-cheating mechanisms, such as tab-switch detection, help ensure that exams are conducted fairly.

Additionally, the system provides clear result displays and performance analytics to help track student progress and identify learning gaps. Results can also be exported as PDF reports for easy sharing and record-keeping. An admin panel enables smooth management of users, exams, and settings, making the entire process organized and user-friendly.

2. Module Mapping

(Provide details of how modules are divided among team members.)

Module Name	Assigned Roll Number(s)	Signature
User Authentication & Profile Management	2520030087	
Exam Builder & Question Management	2520090139	
Online Exam & Timer System	2520030087	
Auto-Grading & Result Management	2520090139	
Analytics, Reports & Notifications	2520030087	

3. Module Details {Student-1: 2520090139 Student-2: 2520030087}

- **Student 1 – Roll No: 2520030087**

- **User Authentication & Profile Management:**

- Develops user registration, login, and profile management features. Handles authentication, user sessions, and secure access for students and administrators.

- **Online Exam & Timer System:**

- Develops the examination interface with timer functionality, answer submission, and anti-cheating measures such as tab-switch detection and activity monitoring.

- **Analytics, Reports & Notifications:**

- Develops performance analytics dashboards, PDF report generation, and notification systems for exam updates and result announcements..

- **Student 2 – Roll No: 2520090139**

- **Exam Builder & Question Management:**

- Implements the module for creating and managing exams with MCQ and essay questions. Designs question categories, exam settings, and question storage.

- **Auto-Grading & Result Management:**

- Implements automatic grading for objective questions and result generation. Displays scores, performance summaries, and exam outcomes.

4. Frameworks / Platforms / Software Used

- **Frontend:** React.js, HTML5, CSS3, JavaScript (ES6+)
- **Styling:** Tailwind CSS
- **Routing:** React Router
- **State Management:** React Context API
- **API Communication:** Axios, REST APIs
- **Backend:** Node.js, Express.js (*optional*)
- **Database:** MongoDB / MySQL (*optional*)
- **Build Tool:** Vite, npm
- **Version Control:** Git, GitHub
- **Deployment:** Vercel / Netlify

Guide Name and Signature (remarks if any) _____